# Materials for Soft Robotic Grippers

## Introduction

Material selection is a critical factor in the design and performance of soft robotic grippers. Unlike rigid grippers, soft grippers require materials that can undergo large and controlled deformation while providing sufficient strength and elasticity for reliable object handling. The material should also maintain its properties over repeated gripping cycles.

For tactile-sensing soft robotic grippers, the material must additionally be compatible with tactile sensors and allow the sensors to detect contact, pressure, or deformation without significantly affecting the flexibility of the gripper. Important considerations include flexibility, elasticity, stiffness, strength, durability, sensor compatibility, and cost.

A wide range of polymer-based materials, including silicone elastomers, TPU, TPE, PU, shape memory polymers, hydrogels, and polymer composites, have been investigated for soft robotic applications. Each material provides a different combination of mechanical and functional properties. Therefore, this chapter focuses on identifying, comparing, and evaluating these materials to determine their suitability for tactile-sensing soft robotic grippers.

## Types of Materials Used in Soft Robotic Grippers

Soft robotic grippers primarily use polymer-based materials because of their flexibility, elasticity, low weight, and ability to undergo large deformations. The major material categories considered for soft grippers are:

### Silicone Elastomers

Silicone elastomers are among the most widely used materials in soft robotic grippers because of their high flexibility, elasticity, and compliance. Common examples include PDMS (Polydimethylsiloxane) and Ecoflex. Their low stiffness allows the gripper to deform easily when interacting with an object, enabling it to conform to objects with different shapes and surfaces. This makes silicone particularly suitable for handling fragile, irregular, or delicate objects.

Silicone elastomers also provide good tear resistance, chemical stability, and resistance to environmental conditions, depending on the formulation. Their elastic deformation allows the gripper to return to its original shape after releasing an object. In tactile-sensing grippers, their soft surface and deformability can be advantageous for integrating or embedding sensing

elements. However, the properties of silicone depend strongly on its formulation and hardness. PDMS is generally more suitable when dimensional stability and sensing integration are important, while softer materials such as Ecoflex provide greater compliance.

**Thermoplastic Elastomers**

Thermoplastic elastomers (TPEs) combine the elastic behavior of rubber with the processing characteristics of thermoplastics. They can undergo significant deformation and recover their original shape after the applied load is removed. This combination makes them useful for flexible robotic structures.

TPU (Thermoplastic Polyurethane) is one of the most commonly investigated TPE-based materials for soft robotic applications. TPU offers a balance of flexibility, toughness, abrasion resistance, and mechanical strength. Unlike conventional silicone elastomers, TPU can be supplied as a filament, making it suitable for extrusion-based 3D printing. Its mechanical properties can also be varied through different grades and hardness levels.

For tactile-sensing grippers, TPU can provide a flexible structure around sensing elements while maintaining sufficient mechanical strength. However, highly flexible TPU can be more difficult to process because of issues such as filament feeding, stringing, and dimensional control during printing.

**Polyurethane (PU)**

Polyurethane (PU) is a versatile polymer whose mechanical properties can be adjusted over a wide range. Depending on its formulation, PU can exhibit flexibility, elasticity, toughness, abrasion resistance, and good fatigue performance. These properties make it useful where the gripper must repeatedly deform while maintaining its structural integrity.

PU can be formulated to obtain different levels of stiffness and flexibility, allowing it to be selected according to the required gripper behavior. Its toughness can also be beneficial for grippers that experience repeated contact with objects. For tactile-sensing applications, PU can serve as a flexible structural material while providing mechanical protection to embedded sensing components.

**Shape Memory Polymers (SMPs)**

Shape memory polymers are stimulus-responsive polymers that can be temporarily deformed and subsequently recover toward their programmed shape when exposed to a suitable stimulus, most

commonly heat. This shape-memory behavior provides an additional functional capability beyond the passive flexibility of conventional elastomers.

SMPs can be designed to produce controlled shape changes, making them attractive for adaptive and reconfigurable soft grippers. Their response can be programmed during material processing, allowing the structure to change its configuration when activated. In a tactile-sensing gripper, SMPs could potentially provide the structural deformation while sensors detect contact or changes in the gripper state.

However, SMPs may have slower response times and temperature-dependent operation, which can limit their suitability for applications requiring rapid and continuous gripping.

**Hydrogels**

Hydrogels are polymeric materials capable of retaining a large amount of water within their three-dimensional network structure. Because of their high water content, they can exhibit very low stiffness and large deformation, giving them mechanical characteristics similar to soft biological tissues.

Their ability to respond to external stimuli such as temperature, pH, light, electric fields, or chemical conditions makes hydrogels interesting for soft robotic applications. Their softness can also make them suitable for applications requiring gentle interaction with objects.

However, hydrogels generally have lower mechanical strength and durability than many elastomers. Water loss, swelling, and environmental sensitivity can also affect their performance. Therefore, although hydrogels are promising for highly deformable and stimulus-responsive systems, their practical application in robust tactile-sensing grippers can be challenging.

**Polymer Composites**

Polymer composites are produced by combining a polymer matrix with reinforcing or functional materials to obtain properties that may not be achievable using a pure polymer. Reinforcements can be used to improve mechanical strength, stiffness, conductivity, or other functional characteristics.

For soft robotic grippers, polymer composites can be designed to create a balance between soft and stiff regions. For example, a flexible polymer matrix can provide deformation while reinforcing elements can control the direction or magnitude of bending. Functional fillers can also introduce electrical or sensing properties, which can be particularly useful for tactile-sensing systems.

The main advantage of polymer composites is their design flexibility, but their fabrication can be more complex than that of single-material polymers. Achieving uniform reinforcement distribution and consistent mechanical properties can also be challenging.

**Material Properties**

The selection of materials for soft robotic grippers depends strongly on their mechanical, physical, and functional properties. Unlike conventional rigid grippers, soft robotic grippers require materials that can undergo large deformation while maintaining sufficient strength and durability. The material must also provide the required flexibility, elasticity, surface interaction, and resistance to repeated loading. Therefore, understanding the important material properties is essential for selecting a suitable material for the gripper.

**Flexibility and Elasticity**

Flexibility is one of the most important properties of a soft robotic gripper because the fingers must bend and deform around objects of different shapes and sizes. Materials with high flexibility can undergo significant deformation without permanent damage. Elasticity allows the material to return toward its original shape after the applied force is removed. Materials such as silicone rubber, TPU, and other elastomers generally provide good flexibility and elasticity, making them suitable for soft gripping applications.

**Tensile Strength**

Tensile strength represents the maximum stress that a material can withstand when it is subjected to a pulling force before failure. A soft gripper needs adequate tensile strength because its fingers experience stretching and internal stresses during operation. A material with insufficient tensile strength may tear or rupture during repeated gripping cycles. Therefore, the material should provide a balance between softness and sufficient tensile strength.

**Hardness**

Hardness indicates the resistance of a material to indentation or deformation. For soft robotic grippers, lower hardness is generally desirable because softer materials can conform more easily to the shape of an object. However, extremely low hardness may reduce the structural stability and load-carrying capability of the gripper. Thus, an appropriate hardness level must be selected according to the intended gripping application.

**Tear Resistance**

Tear resistance is particularly important for soft grippers because thin flexible sections can develop cracks when subjected to repeated deformation. A material with high tear resistance can withstand the propagation of small cuts or defects during operation. This property improves the reliability and service life of the gripper, especially when it is expected to perform a large number of gripping cycles.

**Durability and Fatigue Resistance**

Soft robotic grippers are commonly subjected to repeated bending, stretching, and compression. Therefore, the material must maintain its mechanical properties over a large number of operating cycles. Good fatigue resistance reduces the possibility of permanent deformation, cracking, or failure during prolonged use. Materials with better durability are particularly valuable for industrial and repetitive handling applications.

**Friction and Surface Grip**

The friction between the gripper surface and the object being handled affects the gripping performance. A material with an appropriate coefficient of friction can provide better contact and reduce the possibility of objects slipping from the gripper. Soft elastomeric materials can also increase the contact area by conforming to irregular surfaces. Therefore, surface friction is an important consideration when selecting materials for handling delicate, smooth, or irregularly shaped objects.

**Lightweight Nature**

Low density is desirable in soft robotic grippers because reducing the mass of the gripper decreases the overall load on the robotic system. Lightweight materials can also improve the speed and energy efficiency of robotic movement. Polymers and elastomers generally have much lower density than metals, making them attractive for lightweight soft robotic applications.

**Chemical and Environmental Resistance**

The selected material should be capable of operating under the environmental conditions expected during use. Resistance to moisture, oils, chemicals, temperature variations, and ultraviolet radiation can be important depending on the application. For example, a gripper used in food handling, laboratory environments, or industrial applications may require different levels of chemical and environmental resistance. Hence, environmental stability should also be considered during material selection.

Overall, the ideal material for a soft robotic gripper should provide a suitable combination of flexibility, elasticity, tensile strength, tear resistance, durability, friction, low weight, and environmental resistance. Since no single material is superior in every property, these characteristics must be compared together with material cost and application requirements before making the final selection.

**Material Cost**

The cost of the material is an important factor in the development of soft robotic grippers, particularly for low-cost prototypes, academic research, and applications requiring multiple grippers. Although mechanical and functional performance are primary considerations, selecting an unnecessarily expensive material can significantly increase the overall cost of the system. Therefore, commonly used materials such as silicone elastomers, thermoplastic polyurethane (TPU), thermoplastic elastomers (TPE), and other flexible polymers can be compared based on their approximate material cost, availability, properties, and suitability for soft gripper applications.

| **Material** | **Relative Cost** | **Flexibility** | **Strength** | **Durability** | **Overall Suitability** |
|---|---|---|---|---|---|
| Silicone Elastomer | Medium | Excellent | Moderate | Excellent | Very High |
| TPU | Medium | High | High | High | High |
| TPE | Low-Medium | High | Moderate | Moderate-High | High |
| Flexible Polyurethane | Medium | High | High | High | High |
| Other Flexible Polymers | Low–Medium | Varies | Varies | Varies | Application Dependent |

**Material Selection**

| Selection Criteria | Importance | Preferred Characterstics |
|---|---|---|
| Flexibility | Very High | High flexibility and large deformation |
| Elasticity | Very High | Good recovery after deformation |
| Tensile Strength | High | Sufficient strength to withstand loading |
| Tear Resistance | High | Resistance to crack and tear propagation |
| Durability | High | Good fatigue resistance during repeated cycles |
| Surface Friction | High | Adequate grip and reduced object slipping |
| Weight | Medium | Low density |
| Cost | High | Economical and easily available |
| Overall Suitability | Very High | Balanced performance and cost |

Based on these requirements, silicone elastomer, TPU, and TPE are the most promising materials for soft robotic grippers. The final choice can be made according to the required flexibility, load capacity, durability, manufacturing requirements, and available budget.

# FABRICATION METHODS OF SOFT ROBOTIC GRIPPERS

## Introduction

Fabrication is a critical stage in the development of soft robotic grippers because it determines the gripper's shape, flexibility, strength, and overall performance. Unlike conventional rigid robotic grippers, soft grippers are generally fabricated using flexible materials such as silicone elastomers, TPU, TPE, and other polymers. The fabrication method must be capable of producing complex and deformable structures while maintaining the required dimensional accuracy and mechanical properties.

Several fabrication techniques can be used to manufacture soft robotic grippers, including molding and casting, 3D printing, lamination, and laser-based fabrication. The selection of a suitable method depends on factors such as material compatibility, design complexity, fabrication time, cost, accuracy, and the intended application. Therefore, understanding the advantages and limitations of different fabrication methods is essential for selecting an appropriate technique for soft robotic gripper development.

## 3D Printing

3D printing is an additive manufacturing technique in which a three-dimensional object is produced by depositing or solidifying material layer by layer according to a digital design. It is widely used for soft robotic grippers because it enables the fabrication of complex geometries, customized structures, and lightweight components with relatively little material waste. Depending on the material and printing technology, flexible polymers such as **TPU and TPE** can be used to produce deformable gripper structures.

## Molding and Casting

Molding and casting is one of the most commonly used fabrication methods for soft robotic grippers. In this method, a mold having the desired gripper geometry is prepared, and a liquid or uncured elastomer is introduced into the mold. The material is then allowed to cure and solidify, after which the flexible gripper is removed from the mold. This method is particularly suitable for materials such as silicone elastomers and polyurethane and can produce soft, continuous structures with good flexibility and surface quality.

Molding and casting is relatively simple and economical for producing multiple grippers once the mold has been prepared. However, it may require additional time for mold fabrication and curing, and modifying the gripper design generally requires a new or modified mold.

## Lamination-Based Fabrication

Lamination-based fabrication involves joining multiple layers of flexible materials to form the required structure of a soft robotic gripper. Different layers can be designed to provide specific functions, such as flexibility, reinforcement, or controlled bending. This method is useful for creating thin and lightweight grippers with relatively simple fabrication requirements.

The main advantages of lamination are its low material wastage, simple construction, and ability to combine different materials. However, the bonding between layers must be sufficiently strong to withstand repeated deformation. Poor bonding can lead to delamination and reduce the durability of the gripper.

**Laser Cutting**

Laser cutting is a fabrication technique in which a focused laser beam is used to cut flexible sheets or thin polymer materials according to a predefined digital design. It provides high dimensional accuracy and allows rapid production of relatively simple and planar gripper components. Different layers can also be cut separately and subsequently assembled to form the required soft gripper structure.

The method offers fast fabrication, good repeatability, low material wastage, and easy design modification. However, laser cutting is mainly suitable for sheet-based structures and may not be appropriate for highly three-dimensional or complex internal geometries.

**Comparison of Fabrication Methods**

| **Fabrication Method** | **Flexibility** | **Accuracy** | **Design Complexity** | **Suitability** |
|---|---|---|---|---|
| 3D Printing | High | High | High | High |
| Molding and Casting | Excellent | High | Medium | Very High |
| Lamination | High | Moderate | Low-Medium | High |
| Laser Cutting | Moderate–High | Excellent | Low | High |

**Selection of Fabrication Method**

Based on the comparison of different fabrication techniques, FDM/FFF 3D printing is selected as the most suitable fabrication method for the proposed soft robotic gripper. It enables rapid prototyping, easy modification of the gripper geometry, low material wastage, good repeatability,

and fabrication of complex structures. Compared with conventional fabrication techniques, it also reduces the need for additional molds and allows multiple design iterations to be produced efficiently. Therefore, FDM/FFF 3D printing is selected as the final fabrication method for the proposed soft robotic gripper.